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Project Title: Airline Ticket Reservation Backend System

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1. Introduction

Airline reservation systems are essential for managing flight operations, passenger information and ticket bookings in an organized and efficient manner. With the increasing demand for air travel, airlines must handle large volumes of data accurately and in real time. Traditional or poorly integrated systems often result in inefficiencies such as data inconsistency, booking errors and difficulty in tracking flight and passenger information.

This project focuses on the development of an **Airline Ticket Reservation Backend System** designed to provide a centralized platform for managing flights, passengers and booking operations. The system is implemented using PHP for server-side processing and MySQL for database management, ensuring reliability, scalability and efficient data handling.

The backend system is designed following RESTful API principles, enabling seamless integration with future web or mobile frontend applications. It supports key functionalities such as flight management, passenger data handling, ticket booking and real-time seat availability tracking.

The purpose of this project is not only to solve common challenges in reservation systems but also to provide practical experience in backend development, database design and system architecture. By simulating real-world airline operations, the system serves as a foundation for building scalable and industry-relevant software solutions.

2. Problem Statement

Manual or poorly integrated booking processes create significant operational challenges in managing airline reservations. Without a centralized and well-structured system, it becomes difficult to efficiently organize and track critical information related to passengers, flights and seat availability.

These challenges include:

- Difficulty tracking passenger bookings and flight schedules
- Risk of overbooking or inaccurate seat availability
- Fragmented or inconsistent data storage
- Limited ability to generate analytics for decision-making

As a result, these challenges reduce operational efficiency, increase the likelihood of errors, and make it difficult to maintain reliable and up-to-date reservation data.

To address these problems, a structured backend system using PHP and MySQL will be developed to provide a secure, organized and centralized platform for managing flights, passengers, and bookings efficiently.

3. Objectives

3.1 General Objective:

To develop a robust backend system that efficiently manages airline ticket reservations, including passenger information, flight schedules and booking operations.

3.2 Specific Objectives:

- To Manage flight schedules and route details
- **To Securely store and manage passenger information**
- To Implement booking, reservation and cancellation features
- To Track seat availability in real time
- To Provide admin tools for managing flights and bookings
- To expose RESTful APIs for future integration with web or mobile frontend applications

4. Scope of the Project

The project focuses on developing the backend components of an airline ticket reservation system. The system will include the following functionalities:

- Backend development using **PHP** and **MySQL**
- User authentication and secure login for passengers and administrators
- Flight, passenger and booking management modules
- Real-time Seat availability tracking
- Admin dashboard for monitoring and managing flight and booking data

Excluded Scope

The following features are not included in the current scope of the project:

- Online payment integration
- Mobile application development

However, the system will be designed in a way that allows future integration with web or mobile frontend applications.

5. Significance of the Project

The significance of this project lies in its ability to address common challenges in airline reservation systems such as data inconsistency, overbooking and inefficient tracking. By implementing a centralized backend system, it improves data organization, enhances accuracy, and increases overall operational efficiency.

The system is expected to deliver practical outcomes including a fully functional backend for managing flights, passengers and bookings, along with a secure and well-structured MySQL database. It also provides RESTful API endpoints that enable future integration with web and mobile frontend applications, as well as structured administrative tools for managing airline operations.

From a technical perspective, the project strengthens practical skills in backend development, database design and RESTful API architecture. It provides hands-on experience in building scalable, real-world systems aligned with industry practices.

In addition, the system benefits airline companies by improving operational efficiency, supports students and developers in learning backend and database systems, and assists educational institutions in teaching real-world software engineering concepts. The design also ensures scalability for future enhancements such as mobile applications, payment systems and data analytics modules, making it both a learning platform and a foundation for industry-level solutions.

6. Website Components and Detailed Inclusions

Although the primary focus of this project is backend development, the system is designed to support integration with a web-based frontend interface. The website component will provide user-friendly interaction with the backend services through RESTful APIs.

The system will include the following components:

6.1 Passenger Interface

- Search available flights based on origin, destination and schedule
- View flight details and seat availability
- Enter passenger information and book tickets
- View booking history and status

6.2 Booking and Reservation Module

- Ticket reservation and confirmation system
- Booking cancellation functionality
- Real-time seat availability updates

6.3 Admin Dashboard

- Add, update and delete flight information
- Manage passenger and booking records
- Monitor system activity and booking status
- Control seat allocation and flight schedules

6.4. API Integration Layer

- Communication between frontend and backend using RESTful APIs
- Secure data exchange using structured request and response formats (JSON)

6.5 User Authentication System

- Secure login and registration for passengers and administrators
- Role-based access control (Admin and User roles)

The frontend can be developed using modern technologies such as React, Tailwind CSS and JavaScript/TypeScript to ensure responsiveness, usability and scalability.

7. Methodology

The development of the Airline Ticket Reservation Backend System will follow a structured and systematic approach to ensure efficiency, reliability and scalability. The methodology consists of the following phases:

7.1 Requirement Analysis

In this phase, the system requirements are identified, including user roles (passengers and administrators) and core functionalities such as flight management, booking operations and seat tracking. This stage ensures a clear understanding of system needs before development begins.

7.2 System Design

The system architecture is designed by creating Entity-Relationship (ER) diagrams, database schemas, and defining the API structure. This phase establishes the overall structure of the system and ensures proper data organization and relationships.

7.3 Implementation

The backend system is developed using PHP and MySQL. Key modules such as flight management, passenger management and booking operations are implemented. RESTful APIs are created to enable communication between the backend and future frontend applications.

7.4 Testing

The system is tested to ensure functionality, reliability and data accuracy. Unit testing and API testing are performed using tools such as Postman to validate system performance and identify any issues.

7.5 Documentation

Comprehensive documentation is prepared, including the project report, API documentation and system user guide. This ensures that the system is understandable, maintainable and ready for future enhancements.

7.6 Entity-Relationship Diagram

The Entity-Relationship (ER) diagram illustrates the main entities in the system and the relationships between them. Primary keys (PK) uniquely identify each record, while foreign keys (FK) establish relationships between tables.

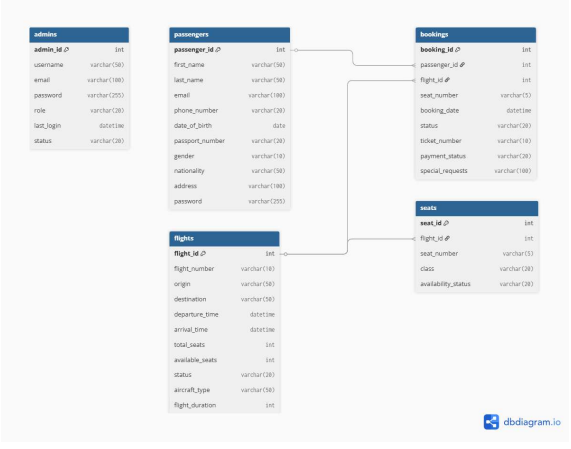
The diagram shows how passengers, flights, bookings, seats and admin accounts interact within the system. This relational structure ensures data consistency and enables efficient management of reservations, passenger information, and flight operations. It also provides a scalable foundation for future integration with frontend applications.

7.7 Database Schema

The database schema defines the main tables used in the airline ticket reservation backend system. Each table stores specific information required for managing passengers, flights, bookings and administrative operations.

Only the key fields are listed below for simplicity. The complete schema with detailed field types and constraints is available online.

Figure 1: ER Diagram for Airline Ticket Reservation Backend System.



Note: The ER diagram visually represents the connections between all main entities, including passengers, flights, bookings, seats and admins.

Table 7.1: Main Database Tables and Key Fields

Table	Key Fields / Example Columns	Description
passengers	passenger_id (PK), first_name, last_name, email, passport_number, phone_number, date_of_birth	Stores passenger info
flights	flight_id (PK), flight_number, origin, destination, departure_time, arrival_time, total_seats, available_seats, status	Stores flight details
bookings	booking_id (PK), passenger_id (FK), flight_id (FK), seat_number, booking_date, status, ticket_number	Stores passenger bookings
seats	seat_id (PK), flight_id (FK), seat_number, class, availability_status	Tracks seats per flight
admins	admin_id (PK), username, email, password, role, last_login, status	Admin accounts and roles

Note: *Full table fields with types are available online: View Full Schema*

8. System Requirements

This section defines the essential requirements needed for the proper development and operation of the Airline Ticket Reservation Backend System.

8.1 Functional Requirements

The system shall provide the following functionalities:

- User registration and secure login for passengers and administrators
- Flight management (add, update, delete and view flights)
- Passenger management and profile handling
- Ticket booking and cancellation system
- Real-time seat availability tracking
- Booking history and status tracking for passengers
- Admin dashboard for managing flights and bookings
- RESTful API endpoints for future frontend integration

8.2 Non-Functional Requirements

The system shall meet the following quality requirements:

- **Security:** Secure authentication and role-based access control for users and admins
- **Performance:** Fast response time for booking and search operations
- **Reliability:** Accurate and consistent data storage without loss
- **Scalability:** Ability to support future expansion such as mobile apps or payment systems
- **Maintainability:** Modular and structured backend code for easy updates
- **Usability:** Simple and well-structured API design for easy integration

8.3 Hardware and Software Requirements

8.3.1 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 20 GB free space
- Internet connection for testing APIs and version control

8.3.2 Software Requirements

Development Tools:

- Visual Studio Code (IDE)
- MySQL Workbench
- Postman (API testing)
- Git and GitHub (version control)

Backend Technologies:

- PHP (server-side development)
- Apache server (XAMPP recommended)

Database:

- MySQL

Frontend Technologies:

- HTML, CSS, JavaScript/TypeScript
- React.js
- Tailwind CSS

9. Use Case

The system interacts with two main actors: **Passenger** and **Admin**.

Passenger Use Cases:

- Register and log in to the system
- Search available flights
- Book tickets
- Cancel bookings
- View booking history

Admin Use Cases:

- Manage flight schedules (add/update/delete flights)
- Manage passenger records
- View all bookings
- Monitor seat availability
- Control system operations

System Behavior:

- The system validates user input before processing
- Updates seat availability automatically after booking or cancellation
- Ensures secure access based on user roles

10. Action Plan and Detailed Work Plan

The project will be developed using a short agile-based approach within **3 to 4 weeks**, divided into focused phases to ensure efficient delivery.

Phase	Key Actions	Deliverables	Responsible Body	Duration	Milestones
Sprint 1: Requirement Analysis	Identify system requirements, user roles, and system features	Requirement specification document	Team Members	3–4 days	Clear understanding of system scope
Sprint 2: System Design	Design ER diagram, database schema, and system architecture	ER diagram, database design	Team Members	4–5 days	Approved system design
Sprint 3: Backend Development	Develop PHP backend, database integration, and core modules	Working backend system	Developers	7–10 days	Functional system modules
Sprint 4: API Development	Create RESTful APIs for system operations	API endpoints (tested in Postman)	Developers	3–4 days	Functional API integration
Sprint 5: Testing	Test system functionality and fix errors	Tested and stable system	Testing team	3–4 days	Bug-free system
Sprint 6: Documentation & Finalization	Prepare final report and documentation	Final project report	All members	2–3 days	Completed documentation

Total Duration: 3–4 Weeks

11. Detailed Budget Breakdown

This section presents an estimated cost of resources used during the development of the system. Since this is an academic project, most tools used are free and the cost is based on development effort and documentation work.

Category	Item/Activities	Description	Estimated Cost (ETB)
Personnel	Backend Development	System design and PHP API development	1500
Personnel	Database Design	MySQL schema design and optimization	1000
Personnel	Testing & Debugging	API testing and error fixing	500
Documentation	Report preparation	Writing and formatting project report	500
Tools	Development tools	VS Code, Git, MySQL Workbench (free tools)	0
Environment	Local server setup	XAMPP / localhost testing environment	0
Miscellaneous	Internet & collaboration	Communication and research support	300

Subtotal: 3800 ETB

Contingency (10%): 380 ETB

Grand Total: 4180 ETB (Estimated)

15. Conclusion

This project aims to deliver a fully functional backend system for airline ticket reservations. By using PHP and MySQL, the system ensures scalability, security and reliability. It addresses common challenges in booking management, such as data inconsistencies, overbooking and manual tracking inefficiencies.

The system also serves as a practical learning platform for students and developers interested in real-world backend development. In the future, it can be extended with features such as payment integration, mobile applications or advanced analytics modules, making it a versatile and industry-ready solution.

16.1 Glossary / Abbreviations

Below are the key terms and abbreviations used in this proposal:

- **PK (Primary Key):** A unique identifier for a record in a database table.
- **FK (Foreign Key):** A field linking one table to another to enforce relational integrity.
- **API (Application Programming Interface):** A set of rules and tools allowing software applications to communicate with each other.
- **Full Database Schema:** All tables, fields and types are available in the file at : gist.github.com/gemachistesfaye/d8bdbd6ed48d14cd3fb82ef0a516703e